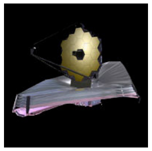

2608 - Revealing the Earliest Reionized Bubbles with JWST

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

In recent years, HST measurements of the rest-frame UV luminosity function have provided a first census of reionization sources, and the efficacy of these sources as Lyman-alpha emitters (LAEs) has been used to constrain the global neutral fraction. Of particular interest are the recent discoveries of bright LAEs at $z > 7$. Empirical evidence suggests that these emitters are being found deep in the reionization epoch, when the IGM was still highly neutral. The observed intensity of Lyman-alpha emission and the occurrence of nearby galaxies has led some to speculate whether these sources reside in cosmological density peaks that were among the first to be reionized. With its enhanced sensitivity and spectroscopic capabilities, the JWST will soon open up an entirely new view of reionization-era galaxies and their environments. Relating the properties of sources to the reionization process around them is a chief goal of high-z galaxy observations. However, since the JWST cannot observe the ionization structure of the IGM directly, a key ingredient for this endeavor is a detailed theoretical understanding of how reionization-era galaxies shaped their cosmological environments and vice versa. The project proposed here aims to provide that understanding. Using an innovative suite of reionization simulations, we propose to study how the spectral properties and clustering of $z > 7$ galaxies are correlated with the structure of the earliest ionized bubbles. The proposed research would guide survey strategies and would help maximize the reionization science yield of JWST.

OBSERVING DESCRIPTION

This work aims to support the reionization science goals of the JWST. NIRCam and NIRSpec data on reionization-era LAEs and LBGs would benefit most from the proposed study.